

A cross-sectional study on the respiratory health status and its determinants among rural handloom workers in Tamil Nadu

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ABSTRACT

Background and Objective: Handloom weaving is a major rural occupation in India, often conducted in home-based settings with limited regulation and minimal protective measures. Such environments may expose workers to chronic airborne dust, leading to adverse respiratory health outcomes. In this context this study aims to assess the respiratory health status of home-based handloom workers in rural Tamil Nadu and identify determinants influencing their respiratory health status. **Methods:** This cross-sectional study included 135 home-based handloom workers selected through multistage sampling. Socio-demographic and occupational data were collected via interviews on a structured proforma. Respiratory health was assessed using the St. George Respiratory Questionnaire (SGRQ). Indoor air quality (PM_{2.5} and PM₁₀) was measured with a calibrated portable monitor. Data were analysed with descriptive statistics and inferential tests, with significance set at $P \leq 0.05$. **Results:** The mean age of participants was 53.4 ± 10.1 years, with 82.2% being male and 48.1% belonging to the lower middle socio-economic class. No participants reported mask usage during work and cross-ventilation was present in 54.8% of workplaces. Based on SGRQ scoring, 50.4% had poor respiratory health status. Age over 50 years, work exposure exceeding 20 years, absence of workplace cross-ventilation, presence of comorbidities, and unhealthy indoor air quality were significant determinants of poor respiratory health ($P < 0.05$). **Conclusion:** A considerable proportion of home-based handloom workers have poor respiratory health status, with no use of personal protective equipment and unhealthy indoor air quality. Interventions to improve ventilation and promoting protective equipment use are essential to safeguard respiratory health in this population.

KEY WORDS: Air pollution, lung diseases, risk factors, rural population, textile industry

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INTRODUCTION

Handloom weaving is centuries-old tradition in India and remains a significant source of rural employment, second only to agriculture in workforce engagement. "Handloom means any loom other than power loom and includes any

hybrid loom on which at least one process of weaving requires manual intervention or human energy for production".^[1] According to the Fourth All-India Handloom Census, there are 31.44 lakh handloom worker households in India,

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reflecting the scale and importance of this sector. Despite its historical and cultural significance, the sector remains largely unregulated and is characterised by traditional practices and lack of occupational health safeguards.^[2]

The textile industry of Tamil Nadu has a significant presence in the national and state economy being a forerunner in industrial development. It is a major contributor towards employment opportunities in the State.^[3] Rural areas, in particular, benefit from the handloom sector, with over 2.09 lakh households and about 2.45 lakh weavers in the state. The majority of handloom weaving households—nearly 88.7%—are concentrated in rural areas, with Tamil Nadu ranking third nationally for the number of handloom weavers.^[2]

Despite its socio-economic importance, handloom weaving is largely unorganised and characterised by informal, home-based working environments. These settings typically lack protective measures and expose workers and their families to diverse health risks, particularly from chronic dust exposure generated during the manufacturing process. Cotton dust is a predominant hazard, causing a range of respiratory ailments including byssinosis, cough, bronchitis, and bronchial asthma.^[4] Bacteria in stored cotton dust, producing endotoxins, have been implicated in the development of byssinosis, a chronic lung disease prevalent among handloom and textile workers.^[5]

Research highlights several risk factors exacerbating health issues among handloom workers, such as the prolonged nature of work, harmful raw materials, poor ventilation, and substandard lighting. Persistent coughs, expectoration, backaches, common colds, and joint pains were found to be the most frequent health complaints among adult weavers.^[6] The combined effect of cotton dust exposure and advancing age is known to lower peak expiratory flow rates, indicating respiratory impairment and the obstructive impact of long-term dust inhalation.^[7] Intensive dust exposure in various textile production stages—spinning, carding, combing—results in different respiratory morbidities, with bacterial endotoxins present in cotton dust identified as a key contributor to lung function impairment and occupational disease.^[8,9]

Given that most home-based handloom operations are family-centred, the hazards also extend to other household members, underscoring the urgent need for targeted research and protective interventions. This study aims to assess the respiratory health status of home-based handloom workers in rural Villupuram district, Tamil Nadu, and to identify the determinants influencing their respiratory health.

MATERIALS AND METHODS

Study design and setting

A community based cross-sectional study was conducted over a period of 3 months (November 2022 to January 2023)

among home-based handloom workers in Villupuram district, Tamil Nadu. Of the 11 blocks in the district engaged in handloom weaving, three blocks (Mugaiyur, Vikravandi, Kandamangalam) with the highest number of handloom households were selected purposively for the study based on data from the Statistics Handbook Villupuram District, 2017–2018.

Sample size

A sample size of 135 was calculated using a prevalence rate of 67.1% for poor respiratory health status,^[10] at a 95% confidence interval and a relative precision of 12%, using the formula $n = Z^2 \times p \times q/d^2$.

Study population and sampling

The study included adults aged over 18 years who had been engaged in home-based handloom weaving for at least 1 year. Individuals not currently engaged in home-based handloom weaving or those unwilling to participate were excluded. The list of home-based handloom worker families in each selected block was obtained from the Office of the Assistant Director of Handlooms and Textiles, Cuddalore. Study participants from each block were selected by systematic random sampling from the list. After obtaining written informed consent, participants were interviewed in their homes during daytime hours to maximise worker inclusion. If an individual did not meet inclusion criteria or was not available at the time of data collection, the next available person was interviewed.

Data collection tools

Data were collected using a pre-tested, interviewer-administered, semi-structured questionnaire. This tool captured socio-demographic details and respiratory health status, utilizing the St. George Respiratory Questionnaire (SGRQ), which is a validated instrument for assessing health status in patients with airway obstruction. The quality of the working environment was evaluated using a calibrated portable instrument designed to measure air pollution levels.

St. George Respiratory Questionnaire (SGRQ)

The SGRQ comprises 50 items covering three components: symptoms (frequency and severity), activities that cause or are limited by breathlessness, and impacts (psycho-social) on daily life. Responses yield domain scores and a total score ranging from 0 to 100, with higher scores indicating worse health status.^[11] The SGRQ shows significant correlations with other indicators of disease activity, including cough, shortness of breath, the 6-minute walk test, and forced expiratory volume in one second (FEV₁), as well as with broader assessments of general health status, such as the Sickness Impact Profile and the 36-Item Short Form Health Survey.^[12] In this study, respiratory health status was classified as “good” or “poor” using the median total SGRQ score as a cutoff.^[13]

Air quality measurement

Indoor air quality was assessed at each participant's work area using a calibrated portable air quality monitor

(Temtop M2000 2nd Generation). The device measures levels of particulate matter (PM_{2.5} and PM₁₀), CO₂, and formaldehyde. Indoor measurements were taken at the centre of the room in which weaving was performed, while outdoor measurements were obtained 100 metres from the house entrance. Air quality index levels were categorised as “acceptable” (Level 1 and 2) or “unhealthy” (Levels 3–6) according to recognised standards [Table 1].^[14–16]

Ethical considerations

The study was approved by the Institutional Ethical Committee of Madras Medical College. Informed written consent was obtained from all study participants, and confidentiality of participant information was strictly maintained.

Data management and statistical analysis

Data were entered in Microsoft Excel and analysed using IBM SPSS version 21. Results are expressed as mean $\pm$ standard deviation for continuous variables and frequencies and percentages for categorical variables. Associations between respiratory health status and potential determinants were analysed using the Chi-square test. A *P* value of ≤ 0.05 was considered statistically significant.

RESULTS

A total of 135 home-based handloom workers participated in this study. The mean age was 53.39 ± 10.08 years, with the majority being male (82.2%). Most participants had attained primary (44.4%) or middle school (31.9%) level education. The majority were married (95.6%) and lived in nuclear families (72.6%). Nearly half (48.1%) belonged to the lower middle-class socio-economic group. Most participants (59.3%) were residing in pucca houses, and over half (54.8%) reported having cross-ventilation in their workplaces. Notably, none of the participants used masks during work. Comorbid conditions were present in 31.1% of the participants, and only 7.4% of participants were smokers [Table 2].

The mean years of occupational exposure among workers was 29.37 ± 10.20 . Indoor and outdoor particulate matter concentrations were measured as given in Table 3.

Respiratory health was assessed using the SGRQ. The total mean score and the mean scores of the three domains are given in Table 4. A total SGRQ score of 25.15 was used as the cut-off: Scores above this indicated poor respiratory

health status. Overall, 50.4% (68) of participants had poor respiratory health according to this criterion [Figure 1].

Table 5 shows the association between respiratory health status and socio-demographic and air-quality variables. Factors found to be significantly associated with poor respiratory health status were age more than 50 (OR = 4.179, 95% CI 2.008–8.695), absence of cross-ventilation in the workplace (OR = 2.712, 95% CI 1.352–5.440), more than 20 years of exposure (OR = 3.424, 95% CI 1.392–8.424), presence of comorbidity (OR = 6.767, 95% CI 2.703–16.939), and unhealthy indoor air quality (OR = 2.259, 95% CI 1.133–4.504).

DISCUSSION

This study assessed the respiratory health status of home-based handloom workers in rural areas of Villupuram district, Tamil Nadu, and identified associated risk factors. The findings revealed that 50.38% of the participants had poor respiratory health as measured by the SGRQ, a tool widely validated for assessing health outcomes in patients with airway obstruction.^[12]

The high prevalence of poor respiratory health among workers aligns with similar studies in the textile industry. A community-based cross-sectional study in Tamil Nadu by Nirmal *et al.*^[17] reported abnormal lung function in 88% of textile workers, with cough and breathlessness as the predominant symptoms. Similarly, Tageldin *et al.*^[18] reported a 60% prevalence of respiratory symptoms among weaving workers in Egypt, and Daivik *et al.*^[10] identified a significant burden of respiratory problems among power loom workers in Tamil Nadu, linked to adverse workplace conditions.

Age was found to be a significant determinant in this study. Participants aged above 50 had over four times higher odds of having poor respiratory health compared to younger workers (OR = 4.179, 95% CI: 2.008–8.695). This aligns with findings by Mishra *et al.*^[19] and Ajeet *et al.*,^[20] who also highlighted that older age groups had increased respiratory morbidity, possibly due to cumulative exposure to dust and declining physiological reserve.

A positive correlation was observed between the duration of exposure and poor respiratory health, with workers exposed for over 20 years showing significantly higher odds of respiratory impairment (OR = 3.424, 95% CI: 1.392–8.424). These findings align with earlier research

Table 1: Air Quality Index (AQI) and level of health concern

AQI Value of Index	Levels of health Concern	PM _{2.5} Conc. $\mu\text{g}/\text{m}^3$	PM ₁₀ Conc. $\mu\text{g}/\text{m}^3$	AQI color	Air Pollution Level
0-50	Good	0-9	0-54	Green	Level 1
51-100	Moderate	9.1-35.4	55-154	Yellow	Level 2
101-150	Unhealthy for sensitive groups	35.5-55.4	155-254	Orange	Level 3
151-200	Unhealthy	55.5-150.4	255-354	Red	Level 4
201-300	Very unhealthy	150.5-250.4	355-424	Purple	Level 5
>301	Hazardous	> 250.5	> 425	Maroon	Level 6

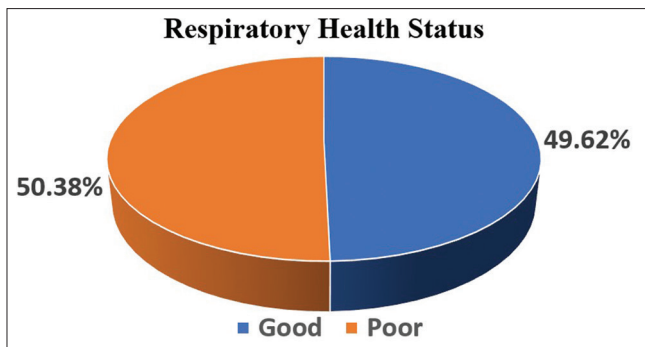


Figure 1: Distribution of respiratory health status among the participants ($n = 135$)

by Ajeet *et al.*^[20] and Fishwick *et al.*,^[21] who reported that prolonged exposure to cotton dust leads to progressive declines in lung function and increases the risk of obstructive lung diseases. Similar results were observed by Sadia *et al.*,^[22] who reported that long-term exposure to cotton dust was significantly associated with elevated rates of asthma, chronic obstructive pulmonary disease, and reduced lung function among textile workers. Similarly, a study from Indonesia by Berlian *et al.*^[23] found that workers exposed to high levels of textile dust had significantly higher odds of respiratory symptoms, with duration of exposure being a key contributing factor. Overall, these studies emphasise the cumulative detrimental impact of long-term occupational exposure on respiratory health.

Environmental factors, particularly indoor air quality and ventilation, played a significant role in determining respiratory health in this study. Workspaces without cross-ventilation showed over twice the odds of poor respiratory health ($OR = 2.71$), emphasising the importance of airflow in mitigating particulate exposure. Similar results were documented by Daba Wami *et al.*,^[24] who found that inadequate ventilation increased respiratory symptoms by 2.4 times in textile workers in Ethiopia. Absence of mechanical ventilation or natural airflow may lead to accumulation of cotton dust and airborne fibres in closed spaces, increasing the risk of inhalation.

Participants' exposure to indoor PM_{2.5}, which exceeded outdoor levels ($36.79 \pm 13.78 \mu\text{g}/\text{m}^3$ vs. $27.79 \pm 2.71 \mu\text{g}/\text{m}^3$), highlights the concerning levels of indoor air pollution within home-based weaving environments. Prior studies have shown that elevated particulate matter concentrations in occupational settings are associated with respiratory morbidity.^[5,25] In the present study, the presence of "unhealthy" indoor air based on standard air quality index classifications was significantly associated with poor SGRQ scores ($OR = 2.259$, 95% CI: 1.133–4.504), emphasising the need for interventions aimed at indoor air quality. Prior research also supports the link between elevated PM_{2.5}/PM₁₀ concentrations, and increased respiratory complaints among textile workers in Southeast Asia.^[23] A similar finding was obtained by Rous *et al.*,^[26]

Table 2: Sociodemographic characteristics of the participants ($n=135$)

Variable	Category	<i>n</i> (%)
Gender	Male	111 (82.2)
	Female	24 (17.8)
Education	Illiterate	14 (10.4)
	Primary School	60 (44.4)
	Middle School	43 (31.9)
	High School	17 (12.6)
	Degree	1 (0.7)
Marital Status	Married	129 (95.6)
	Widowed	6 (4.4)
Family Type	Nuclear	98 (72.6)
	Joint	37 (27.4)
Socio-economic status	Upper Middle	3 (2.2)
	Middle	44 (32.6)
	Lower Middle	65 (48.1)
	Lower	23 (17)
House Type	Pucca	80 (59.3)
	Semipucca	55 (40.7)
Cross-Ventilation	Present	74 (54.8)
	Absent	61 (45.2)
Wearing a Mask While working	Yes	0 (0)
	No	135 (100)
Co-morbidities	Present	42 (31.1)
	Absent	93 (68.9)
Smoking	Yes	10 (7.4)
	No	125 (92.6)

Table 3: Air quality parameters

Parameter	Mean (SD)	Level of health concern
Outdoor PM 2.5	27.79 (2.707)	Moderate
Outdoor PM 10	31.47 (3.314)	Good
Indoor PM 2.5	36.79 (13.78)	Unhealthy
Indoor PM 10	42.97 (16.415)	Good

Table 4: Mean domain scores and total score of SGRQ

SGRQ Domain	Mean±SD
Symptoms	42.17±19.34
Activity	34.31±21.39
Impact	26.48±22.72
Total score	31.66±19.74

where workers exposed to high organic dust levels had statistically significant rises in SGRQ scores.

These findings suggest that indoor air quality is a modifiable environmental determinant of respiratory morbidity among home-based handloom workers. Interventions should prioritise improving structural ventilation, reducing dust generation at the source, and increasing community awareness of indoor air hazards. Implementing low-cost air filtration methods along with periodic monitoring of indoor air quality could serve as practical starting points in resource-limited settings.

One of the most striking findings was the absence of use of personal protective equipment (PPE) by the participants. Despite well-documented links between dust exposure and chronic respiratory disease,^[4,7] none of the participants reported the use of masks or other protective items. This gap highlights the need for occupational health training,

Table 5: Factors associated with respiratory health status (n=135)

Variables		Respiratory health Status		P	Odds Ratio (95% CI)
		Good n (%)	Poor n (%)		
Age	<50	39 (69.6)	17 (30.4)	0.001*	4.179 (2.008–8.695)
	>50	28 (35.4)	51 (64.6)		
Education	Up to primary school	36 (48.6)	38 (51.4)	0.802	0.917 (0.465–1.806)
	Middle school and above	31 (50.8)	30 (49.2)		
Family type	Nuclear	49 (50)	49 (50)	0.889	1.056 (0.495–2.249)
	Joint	18 (48.6)	19 (51.4)		
House type	Pucca	35 (43.8)	45 (56.3)	0.099	0.559 (0.279–1.119)
	Semipucca	32 (58.2)	23 (41.8)		
Years of Exposure	<20	21 (72.4)	8 (27.6)	0.006*	3.424 (1.392–8.424)
	>20	46 (43.4)	60 (56.6)		
Cross-Ventilation	Present	41 (62.1)	25 (37.9)	0.005*	2.712 (1.352–5.440)
	Absent	26 (37.7)	43 (62.3)		
Monthly income	<6500	40 (44.9)	49 (55.1)	0.130	0.574 (0.280–1.181)
	>6500	27 (58.7)	19 (41.3)		
Smoking	Present	3 (30)	7 (70)	0.325	0.408 (0.101–1.652)
	Absent	64 (51.2)	61 (48.8)		
Comorbidity	Absent	60 (61.2)	38 (38.8)	0.001*	6.767 (2.703–16.939)
	Present	7 (18.9)	30 (81.1)		
Indoor air quality	Acceptable	42 (59.2)	29 (40.8)	0.020*	2.259 (1.133–4.504)
	Unhealthy	25 (39.1)	39 (60.9)		

*Statistically significant at $P < 0.05$ by Chi square test

behavioural intervention, and access to protective tools. Overall, these findings represent a complex interplay between individual characteristics, work exposure, and environmental exposures in the development of respiratory disease in handloom weavers. The results are in concordance with global literature on dust-related occupational lung diseases such as byssinosis and chronic obstructive pulmonary disorders, often reported in unregulated textile sectors in low- and middle-income countries.^[4,9,27]

Strengths and limitations

To our knowledge, this is among the few studies focussing on home-based handloom workers—a population often excluded from formal occupational health research. The use of a validated tool (SGRQ) and calibrated air quality monitoring enhances the reliability of the findings. However, there are limitations. Lung function was not objectively measured using spirometry, and the cross-sectional design limits the ability to establish causality.

CONCLUSION

This cross-sectional study highlights a significant burden of poor respiratory health among home-based handloom workers in the rural settings of Villupuram district, Tamil Nadu. Over half of the study participants had poor respiratory health status based on validated SGRQ scores. A few of the risk factors that significantly correlated with poor respiratory health status were age above 50 years, exposure of longer than 20 years to weaving, absence of cross-ventilation, pre-existing comorbidities, and exposure to unhealthy indoor air quality. Notably no worker indicated the use of masks or other personal protective gear at work, reflecting critical gaps in occupational health practice.

RECOMMENDATIONS

Given these findings, multifaceted interventions to improve the health and safety of handloom workers is essential. Future efforts should also include longitudinal research employing objective clinical measures such as spirometry, alongside targeted health education to promote awareness and appropriate use of PPE. Changes in infrastructure to ensure adequate cross-ventilation and community-based monitoring of ambient air quality are crucial for sustaining improvements. Collaboration with policy makers and integration of occupational health surveillance into broader public health programs may significantly aid in early detection and prevention of respiratory illnesses.

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Conflicts of interest

There are no conflicts of interest.

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